INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE NAME

CLASS

INDEX NUMBER

CHEMISTRY

8872/02

Section A: Structured Questions
Section B: Free Response Questions

2012

2 hours

Candidates answer on the Question Paper
Additional Materials: *Data Booklet*

READ THESE INSTRUCTIONS FIRST

Write your index number, name and civics group.
Write in dark blue or black pen.
You may use pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the space provided.
A Data Booklet is provided.

You are advised to show all working in calculations.
You are reminded of the need for good English and clear presentation in your answers.
You are reminded of the need for good handwriting.
Your final answers should be in 3 significant figures.

You may use a calculator.

The number of marks is given in brackets [ ] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	11
2	9
3	7
4	13
Significant figures	
Handwriting	
Total	40

This document consists of **11** printed pages.

For Examiner's Use

Section A

Answer ALL questions on the spaces provided.

1. **(a)** The elements in Period 3 burns with fluorine to produce the corresponding fluorides.

The formulae and melting points of the fluorides of the elements in Period 3, Na to C*l* are given in the table.

Formula of fluoride	NaF	MgF_2	AlF_3	SiF_4	PF_5	SF_6	ClF_5
m.p. / K	1268	990	1017	183	139	223	170

(i) Describe the bonding in the compound sodium fluoride. Draw a diagram to illustrate your answer.

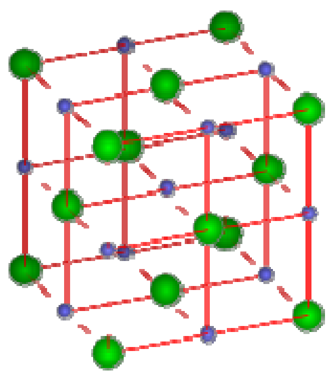
……………………………………………………………………………………

……………………………………………………………………………………

……………………………………………………………………………………

……………………………………………………………………………………[2]

sodium fluoride has a giant ionic structure with strong electrostatic attraction between the oppositely charged cations and anions.



(ii) Complete the electronic configuration for the following species.

the A*l* atom $1s^2$……………………………………………………

the F^- ion $1s^2$……………………………………………………

[2]

For Examiner's Use

the A*l* atom : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$
the F^- ion : $1s^2\ 2s^2\ 2p^6$

(b) The elements burnt in oxygen to produce oxides.

(i) Describe what would be observed when sulfur reacts with oxygen.

....

....[1]

Sulfur burns in oxygen on gentle heating with a brilliant blue flame.

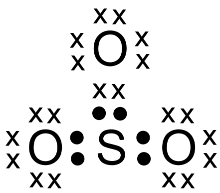
(ii) Write a balanced equation, including state symbols, for the reaction in **(i)**.

....[1]

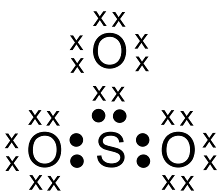
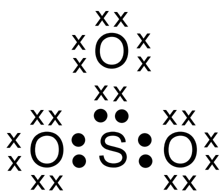
$S(s) + O_2\,(g) \rightarrow SO_2\,(g)$
or
$S_8(s) + 8O_2\,(g) \rightarrow 8SO_2\,(g)$

(iii) The oxide formed in **(ii)** can be further oxidised to SO_3 by adding an appropriate catalyst, V_2O_5.

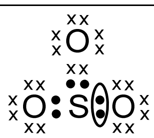
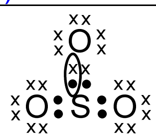
A 'dot-and-cross' diagram of a SO_3 molecule is shown below. Only electrons from outer shells are represented.



In the table below, there are two copies of this structure.
On the structures, draw a circle around a pair of electrons that is associated with **each** of the following.

(I) A co-ordinate bond	(II) a covalent bond
	

[1]

(I) A co-ordiate bond	(II) a covalent bond
	 Circle only a pair of electrons

(c) An unstable isotope phosphorus-35, ^{35}P, produces sulfur-35, ^{35}S when a neutron is converted to a proton. This process is known as beta emission.

For Examiner's Use

(i) Deduce the number of protons and neutrons isotope sulfur-35, ^{35}S contains.

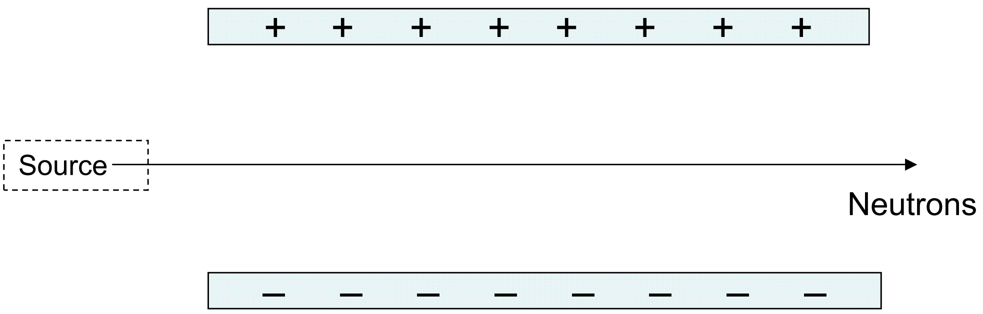
protons ………………………..

neutrons ……………………… [2]

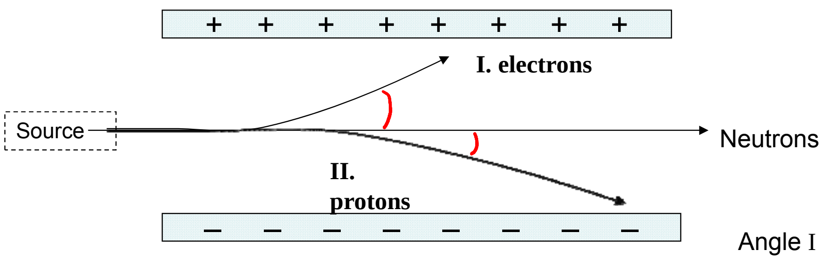
Protons: 16
Neutrons: 19

(ii) Sketch on the diagram below how beams of each of the following particles are affected by the electric field, labelling each beam clearly.

I. electrons
II. protons



[2]

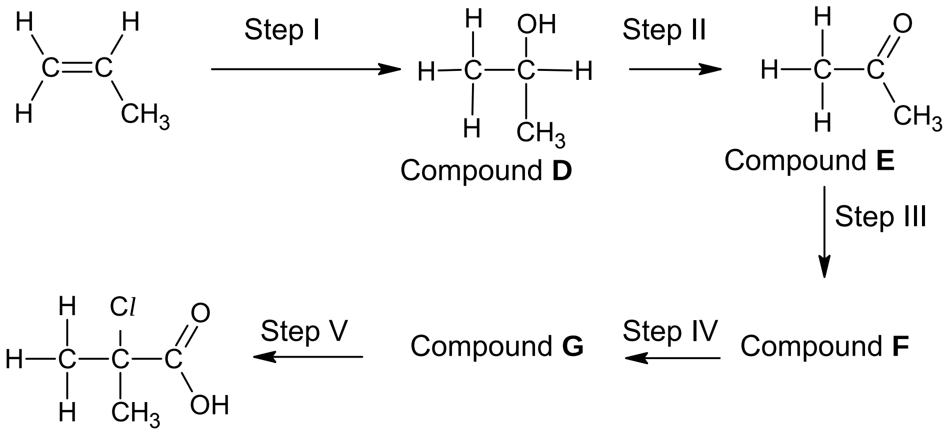


Angle I
>angle II

[Total:11]

For Examiner's Use

2 Prop-1-ene is an important raw material for the production of polypropylene, a recyclable polymer commonly used to make plastic containers. Compound **I** can be produced from prop-1-ene in the following synthesis pathway.



Compound **H**

(a) Suggest reagents and conditions for steps I and II.

Step I

Reagent and Condition:...

Step II

Reagent and Condition:...

[2]

Step I
steam, H_3PO_4, 60 atm, 300^oC
OR concentrated H_2SO_4 followed by steam

Step II
$KMnO_4$(aq),H_2SO_4(aq), heat with reflux
OR $K_2Cr_2O_7$(aq),H_2SO_4(aq), heat with reflux

(b) Propose a three-step synthesis for the conversion of Compound **E** to Compound **H**, giving the structural formulae of the intermediates, Compound **F** and Compound **G** in the space provided.

Step III

Reagent and Condition:...

Step IV

Reagent and Condition:...

Step V

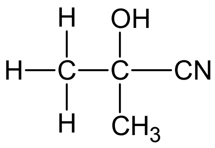
Reagent and Condition:...

For Examiner's Use

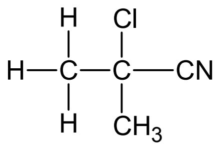
Stage III : HCN, trace NaCN/NaOH(aq), 10°C – 20°C

Stage IV : PCl_5, room temperature

Stage V : Dilute H_2SO_4/HCl, heat with reflux



Compound G



Compound H

Compound **G**

Compound **F**

(c) Suggest a simple chemical test to distinguish Compound **D** and propanol. State what would be observed during the test.

Reagent and Condition:...

Compound **D**:...

Propanol:...

[2]

Reagent and condition: aq Iodine, aq sodium hydroxide,warm

Compound E: yellow ppt of CHI_3 seen

Propanol: no yellow ppt seen.

[Total: 9]

3 **(a)** In theory, the reaction shown below could be used to prepare SiF_4.

$$CF_4(g) + SiH_4(g) \longrightarrow CH_4(g) + SiF_4(g)$$

The table below shows bond enthalpy data for covalent bond in its gaseous phase.

Gas-phase bond	Average bond enthalpy / kJ mol^{-1}
C–H	413
C–F	467
Si–H	318
Si–F	553

Calculate the enthalpy change of the reaction, including the units, shown in the given equation.

[2]

For Examiner's Use

enthalpy change of the reaction
= [4 x 467 + 4 x 318] – [4 x 413 + 4 x 553]
= **-724 kJ mol^{-1}**

(b) The reaction shown in **(a)** is not observed to take place at room temperature and pressure. However, recent research has revealed a method of exchanging a fluorine atom bonded to carbon with a hydrogen atom bonded to silicon at room temperature and pressure. This was accomplished by introducing triethylsilyl cations, $(C_2H_5)_3Si^+$. The proposed mechanism contains the following steps.

Step 1 $(C_2H_5)_3Si^+ + C_6H_5CF_3 \longrightarrow (C_2H_5)_3SiF + C_6H_5CF_2^+$
Step 2 $(C_2H_5)_3SiH + C_6H_5CF_2^+ \longrightarrow (C_2H_5)_3Si^+ + C_6H_5CF_2H$

(i) Write down the overall equation shown by the reaction steps above.

[1]

$(C_2H_5)_3SiH + C_6H_5CF_3 \rightarrow (C_2H_5)_3SiF + C_6H_5CF_2H$

(ii) State and explain the role of $(C_2H_5)_3Si^+$ in this process.

[2]

A **catalyst**

It **accepts** a fluoride ion and is **regenerated** in the following step,
Or
It **lowers the activation energy** of the reaction by **providing an alternative pathway**.

(c) Suggest why Si–F bonds have a higher average bond enthalpy than C–F bonds.

[1]

Si-F bond is more polar / greater electronegativity difference/ **covalent bond with some ionic character**

(d) Suggest why, despite the stronger Si–F bond, SiF_4 is more reactive than CF_4.

[1]

SiF_4 is more reactive than CF_4 because of **lower activation energies** in the mechanism.
OR
Si can expand octet using its energetically accessible d-orbitals & can participate in the reaction with SiF_4 (which creates the potential for mechanisms with lower E_a)

OR
Carbon being smaller size has steric hinderance around it.

[Total: 7]

For Examiner's Use

4 (a) Carbon monoxide, which can be used to make methanol, may be formed by reacting carbon dioxide with hydrogen.

$$CO_2(g) + H_2(g) \rightleftharpoons CO(g) + H_2O(g) \qquad K_c = 1.44 \text{ at } 1200\text{ K}$$

(i) Write K_c expression for the above reaction, stating its units.

[1]

$K_c = \dfrac{[CO]\,[H_2O]}{[H_2][CO_2]}$ No units

(ii) A mixture containing 0.50 mol of CO_2, 0.50 mol of H_2, 0.20 mol of CO and 0.20 mol of H_2O was placed in a 2.0 dm^3 flask and allowed to come to equilibrium at 1200 K.
Calculate the amount, in moles, of each substance present in the equilibrium mixture at 1200 K.

	CO_2	+	H_2	$\rightleftharpoons$	CO	+	H_2O
Initial mol	0.50		0.5		0.20		0.20

[5]

	CO_2	+	H_2	$\rightleftharpoons$	CO	+	H_2O
Initial mol	0.50		0.5		0.20		0.20
Change in mol	-x		-x		+x		+x
Eqm mol	0.5 –x		0.5 –x		0.2 + x		0.2 + x

$K_c = \dfrac{[CO]\,[H_2O]}{[H_2][CO_2]}$

$= \dfrac{(0.2+x)^2}{(0.5-x)^2} = 1.44$

Take root on both sides, and solve,.
X = 0.182

At eqm,
Mol of CO = mol of H_2O = **<u>0.382</u>**

Mol of CO_2 = mol of H_2 = **<u>0.318</u>**

(b) (i) Write an equation for the dissociation of water.

[1]

$H_2O \rightleftharpoons H^+ + OH^-$

For Examiner's Use

(ii) Use the equation in **(i)** to write an expression for the equilibrium constant, K_c, for this reaction.

$$K_c = \frac{[H^+][OH^-]}{[H_2O]}$$

(iii) Use the expression in **(ii)** to show that $K_w = [H^+][OH^-]$. Justify and explain your reasoning.

[2]

Since **water is in large excess** & **relatively constant concentration**,and
$K_c [H_2O] = [H^+][OH^-]$
$K_w = [H^+][OH^-]$ Where $K_w = K_c [H_2O]$ is a constant

(iv) At 373 K, the ionic product of water, K_w, has a value of 51.3×10^{-14} mol^2 dm^{-6}. Use this information to calculate the pH of water at 373 K. Give your answer to 3 significant figures.

[1]

$K_w = [H^+][OH^-]$
$[H^+] = \sqrt{(51.3 \times 10^{-14})} = 7.16 \times 10^{-7}$
pH = **6.15**

(iv) At 298 K the pH of pure water is measured to be 7.0. When pure water is warmed to 338 K, it is observed that its pH drops to 6.7.
Find the pK_w of pure water at 338 K.

[1]

At 338 K, pK_w = pH + pOH = 2 x 6.7 = **13.4**

(v) Using the information in **(iv),** calculate the pH of 0.100 mol dm^{-3} aqueous sodium hydroxide at 338K.

[1]

$[OH^-]$ = 0.100 mol dm^{-3}
pOH = - lg [0.100] = 1
pH = 13.4 – 1 = **12.4**

[Total: 13]